

Linear Algebra

Dr. Harshitha A.

Inner Product

Let u and v be any two vectors in $\mathbb{R}^n$. That is., $u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$ and $v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$.

Then the inner product of u and v is,

$$\begin{aligned} u \cdot v &= u^T v = \begin{bmatrix} u_1 & u_2 & \dots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \\ &= u_1 v_1 + u_2 v_2 + \dots + u_n v_n \end{aligned}$$

Example

Compute $u \cdot v$ and $v \cdot u$ for $u = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$ and $v = \begin{bmatrix} 3 \\ 2 \\ -3 \end{bmatrix}$

Let u , v and w be vectors in $\mathbb{R}^n$, and c be a scalar. Then

- ① $u \cdot v = v \cdot u$
- ② $(u + v) \cdot w = u \cdot w + v \cdot w$
- ③ $(cu) \cdot v = c(u \cdot v) = u \cdot (cv)$
- ④ $u \cdot u \geq 0$ and $u \cdot u = 0$ if and only if $u = 0$

The length(or norm) of a vector

The length (or norm) of v in $\mathbb{R}^n$ is the non-negative scalar $\|v\|$ defined by

$$\|v\| = \sqrt{v \cdot v} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

- ① Length in $\mathbb{R}^2$
- ② Length in $\mathbb{R}^3$

For any scalar, c ,

$$\|cv\| = |c| \|v\|$$

Unit vector

A vector whose length is 1 is called the unit vector.

Example

$$v = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

Normalizing a vector

If we divide a non-zero v vector by its norm $\|v\|$, then the resulting vector is unit vector in the direction of v .

Example

$$v = \begin{bmatrix} 1 \\ -2 \\ 2 \\ 0 \end{bmatrix}$$

Distance in $\mathbb{R}^n$

Let $u, v \in \mathbb{R}^n$. The distance between u and v is $\text{dist}(u, v) = \|u - v\|$

Example

1. Compute the distance between the vectors $u = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Example

2. Compute the distance between the vectors $u = \begin{bmatrix} 7 \\ 1 \\ -1 \end{bmatrix}$ and $v = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}$

Orthogonal vectors

Two vectors u and v in $\mathbb{R}^n$ are orthogonal if $u \cdot v = 0$

Example

Check whether $u = \begin{bmatrix} \frac{4}{3} \\ -1 \\ \frac{2}{3} \end{bmatrix}$ and $v = \begin{bmatrix} 5 \\ 6 \\ -1 \end{bmatrix}$ are orthogonal vectors.

Problems

1. Let $u = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $v = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$. Then find

- a $u \cdot v$
- b $\frac{u \cdot v}{v \cdot v} v$
- c $\frac{u}{||u||^2}$

2. Find unit vector in the direction of the following vectors.

a $u = \begin{bmatrix} -30 \\ 40 \end{bmatrix}$

b $u = \begin{bmatrix} \frac{7}{4} \\ \frac{1}{2} \\ 1 \end{bmatrix}$

3. Find the distance between $u = \begin{bmatrix} 0 \\ -5 \\ 2 \end{bmatrix}$ and $u = \begin{bmatrix} -4 \\ -1 \\ 8 \end{bmatrix}$

4. Determine which pair of vectors are orthogonal

a $a = \begin{bmatrix} 8 \\ -5 \end{bmatrix}$ and $b = \begin{bmatrix} -2 \\ -3 \end{bmatrix}$

4. Determine which pair of vectors are orthogonal

a $a = \begin{bmatrix} 8 \\ -5 \end{bmatrix}$ and $b = \begin{bmatrix} -2 \\ -3 \end{bmatrix}$

b $y = \begin{bmatrix} 3 \\ 2 \\ -5 \\ 0 \end{bmatrix}$ and $z = \begin{bmatrix} -4 \\ 1 \\ -2 \\ 6 \end{bmatrix}$

Orthogonal sets

A set of vectors $\{u_1, u_2, \dots, u_p\}$ in $\mathbb{R}^n$ are orthogonal if each pair of distinct vectors in the set are orthogonal.

Example

$u_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$ and $u_3 = \begin{bmatrix} -\frac{1}{2} \\ -2 \\ \frac{7}{2} \end{bmatrix}$ are orthogonal.

Properties of orthogonal vectors

Theorem

If $S = \{u_1, u_2, \dots, u_p\}$ is an orthogonal set of non-zero vectors in $\mathbb{R}^n$, then S is linearly independent and hence is a basis for the subspace spanned by S .

Orthonormal vectors

Definition

Two vectors u and v in $\mathbb{R}^n$ are orthonormal if $u \cdot v = 0$ and $\|u\| = \|v\| = 1$.

Example

$$u = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \text{ and } v = \begin{bmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

Orthonormal sets

A set of vectors $\{u_1, u_2, \dots, u_p\}$ in $\mathbb{R}^n$ are orthonormal if each pair of distinct vectors in the set are orthonormal.

Theorem

If $S = \{u_1, u_2, \dots, u_p\}$ is an orthonormal set of non-zero vectors in $\mathbb{R}^n$, then S is linearly independent and hence is a basis for the subspace spanned by S .

Example

Show that $\{v_1, v_2, v_3\}$ is an orthonormal basis of $\mathbb{R}^3$, where $v_1 = \begin{bmatrix} \frac{3}{\sqrt{11}} \\ \frac{1}{\sqrt{11}} \\ \frac{1}{\sqrt{11}} \end{bmatrix}$,

$$v_2 = \begin{bmatrix} \frac{-1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix} \text{ and } v_3 = \begin{bmatrix} \frac{-1}{\sqrt{66}} \\ \frac{-4}{\sqrt{66}} \\ \frac{7}{\sqrt{66}} \end{bmatrix}$$

Theorem

An $m \times n$ matrix U has orthonormal column if and only if $U^T U = I$. If a matrix has orthonormal columns, then the matrix is **orthonormal matrix**.

Example

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{2}{3} \\ \frac{1}{\sqrt{2}} & -\frac{2}{3} \\ 0 & \frac{1}{3} \end{bmatrix}$$

The Gram-Schmidt Process

The Gram-Schmidt Process is a simple algorithm for producing an orthogonal or orthonormal basis for any non-zero subspace $\mathbb{R}^n$

Given a basis $\{x_1, x_2, \dots, x_p\}$ for a nonzero subspace W of $\mathbb{R}^n$, define

$$v_1 = x_1$$

$$v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1$$

$$v_3 = x_3 - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} v_2$$

$$\vdots$$

$$v_p = x_p - \frac{x_p \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_p \cdot v_2}{v_2 \cdot v_2} v_2 - \dots - \frac{x_p \cdot v_{p-1}}{v_{p-1} \cdot v_{p-1}} v_{p-1}$$

Then, $\{v_1, v_2, \dots, v_p\}$ is an **orthogonal basis** for W . An **orthonormal basis** is constructed from an orthogonal basis $\{v_1, v_2, \dots, v_p\}$ by normalizing all v_k , $1 \leq k \leq p$.

Example

1. Let $W = \text{Span}\{x_1, x_2\}$, where $x_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$ and $x_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Construct an orthogonal and orthonormal basis for W .

Example

2. Let $W = \text{Span} \{x_1, x_2, x_3\}$, where $x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ and $x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$.

Construct an orthogonal and orthonormal basis for W .

QR Factorization

If A is an $m \times n$ matrix with linearly independent columns, then A can be factored as $A = QR$, where

- Q is an $m \times n$ matrix whose columns form an orthonormal basis for $\text{Col } A$
- R is an $n \times n$ triangular invertible matrix with positive entries on its diagonal.

- Let $A = [x_1 \ x_2 \ \dots \ x_n]$, where each x_i is a vector in $\mathbb{R}^m$.
- Find orthogonal vectors $v_1, v_2, \dots v_n$.
- Find orthonormal vectors $u_1, u_2, \dots u_n$.
- The matrix Q is $Q = [u_1 \ u_2 \ \dots \ u_n]$
- We know $A = QR$. Consider

$$\begin{aligned} Q^T A &= Q^T (QR) \\ &= (Q^T Q) R \\ &= IR = R \end{aligned}$$

- Therefore the matrix R can be obtained by $R = Q^T A$

Example

1. Find the QR factorization of $A = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix}$

Example

2. Find the QR factorization of $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Singular Value Decomposition

Let A be an $m \times n$ matrix. Then $A^T A$ is a square symmetric matrix and can be orthogonally diagonalized.

Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be eigenvalues of $A^T A$. Note that $\lambda_1, \lambda_2, \dots, \lambda_n$ are non-negative. We can arrange the eigenvalues as $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$, where λ_1 is the largest eigenvalue of $A^T A$. The singular values of A are the square roots of the eigenvalues of $A^T A$, denoted as $\sigma_1, \sigma_2, \dots, \sigma_n$.

Example

Find singular values of the matrix $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$

The singular value decomposition

The decomposition of A involves an $m \times n$ diagonal matrix Σ of the form $\Sigma = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix}$, where D is $r \times r$ diagonal matrix for some r not exceeding the smaller of m and n .

The singular value decomposition theorem

Let A be an $m \times n$ matrix with rank r . Then there exists an $m \times n$ matrix Σ for which the diagonal entries in D are the first r singular values of A , $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r \geq 0$ and there exists an $m \times m$ orthogonal matrix U and the $n \times n$ orthogonal matrix V such that $A = U\Sigma V^T$

1. Construct singular value decomposition of $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$

Steps:

- ① Find eigenvalues $A^T A$
- ② Find eigenvectors corresponding to each eigenvalues, say $v_1, v_2, \dots, v_n$
- ③ Normalize $v_1, v_2, \dots, v_n$ to get $u_1, u_2, \dots, u_n$
- ④ Form matrix $V = \begin{bmatrix} u_1 & u_2 & \dots & u_n \end{bmatrix}$
- ⑤ Find singular values of A
- ⑥ Form matrix D ($r \times r$) and Σ ($m \times n$)
- ⑦ The matrix $U = \begin{bmatrix} Au_1 & Au_2 & \dots & Au_r \end{bmatrix}$
- ⑧ $A = U\Sigma V^T$